Direct Preference Optimization: From Paraphrase Detection to Shakespearean Sonnet Generation

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Problem

Aligning language models with human preferences has become a central challenge in NLP. While **Reinforcement Learning from Human Feedback (RLHF)** achieves strong results, its complexity, requiring simultaneous management of multiple models and costly online sampling, motivates simpler alternatives.

Direct Preference Optimization (DPO) addresses this by reformulating alignment as a classification problem, eliminating the need for an explicit reward model. However, DPO's effectiveness has been evaluated almost exclusively on open-ended generation tasks.

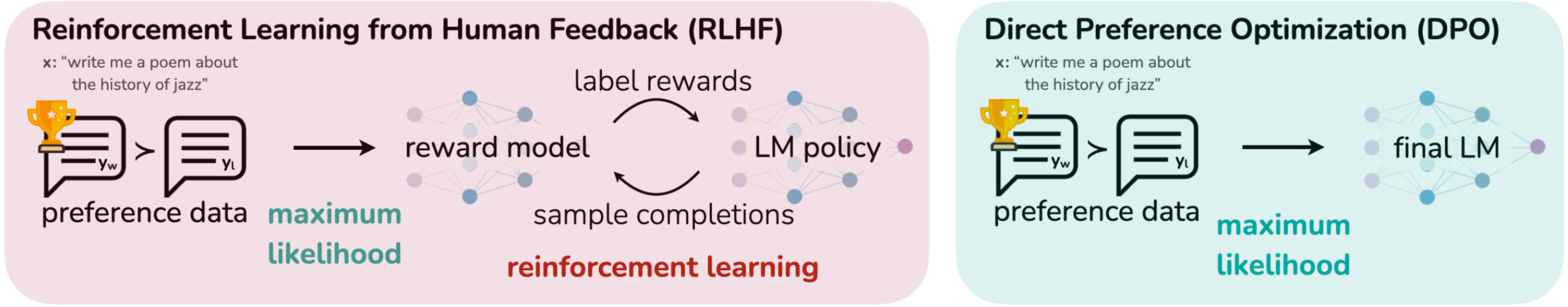

Figure 1. DPO optimizes for human preferences while avoiding reinforcement learning [?].

Goal: This work presents a comprehensive empirical evaluation of DPO applied to two underexplored tasks: paraphrase detection and Shakespearean sonnet generation, both using fine-tuned GPT-2 base models as starting points.

Background

These tasks introduce distinct challenges:

- **Paraphrase detection** offers limited room for improvement given strong supervised baselines.
- **Sonnet generation** demands strict structural and stylistic adherence, which is inherently subjective to evaluate.

DPO Formulation

DPO optimizes the policy directly on preference-labeled datasets using a binary cross-entropy loss:

$$L_{DPO}(\pi_\theta; \pi_{ref}) = -\mathbb{E}_{(x,y_w,y_l)\sim D}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y_w\mid x)}{\pi_{ref}(y_w\mid x)} - \beta\log\frac{\pi_\theta(y_l\mid x)}{\pi_{ref}(y_l\mid x)}\right)\right]$$

Where:

- σ is the sigmoid function
- π_θ represents the model being optimized
- π_{ref} is a fixed pretrained or fine-tuned base model
- β is a temperature parameter controlling the strength of the preference signal
- (x, y_w, y_l) represents a prompt and its preferred (y_w)/dispreferred (y_l) output pair

Methods

The research methodology is organized into a sequential two phases starting from a fine-tuned GPT-2 base model.

Phase I: Supervised Fine-Tuning (SFT)

1. Full-parameter fine-tuning on the Quora Question Pairs dataset for paraphrase detection.
2. Causal language modeling fine-tuning on a Shakespearean sonnets dataset.

Phase II: Direct Preference Optimization (DPO)

- **Data Preparation**: Gemini 2.5 Flash acted as a high-capacity teacher model to generate preference labels for both tasks.
- **Model Optimization**: Fine-tune the initial SFT policy π_θ by minimizing the L_{DPO} loss function, keeping the starting checkpoint as the reference model π_{ref} to prevent distributional drift.

Experiments

Paraphrase Identification

- **Inputs**: Sentence pair. **Outputs**: Paraphrase classification (Binary).
- **Results**: Consistent, modest improvements. Test accuracy improved from 0.840 to 0.850. Validation metrics similarly showed marginal gains over the strong SFT baseline.

	Validation Dataset				Test Dataset
	Accuracy	**F1**	**Precision**	**Recall**	**Accuracy**
Baseline (SFT)	0.897	0.892	0.887	0.899	0.840
DPO Refined	0.899	0.892	0.890	0.895	**0.850**

Table 1. Evaluation results for Paraphrase detection.

Sonnet Generation

- **Inputs**: 3 initial lines. **Outputs**: 11 remaining lines completing the sonnet.
- **Results**: Quantitative **chrF** scores (character n-gram F-score) showed mixed results. It increased from 41.543 to 42.690 on validation but slightly decreased from 41.890 to 41.622 on the test set.

Analysis

A qualitative analysis of the generated sonnets confirms that the text produced after applying DPO is **substantially better** across multiple dimensions invisible to automatic n-gram metrics like chrF.

Structural and Stylistic Fidelity

- **14-Line Target:** The DPO model strictly hits the 14-line sonnet target and avoids catastrophic generation collapse.
- **Rhyme Schemes:** While neither model perfected ABAB CDCD EFEF GG, the baseline failed completely whereas DPO makes genuine *slant-rhyme* attempts in continuations.
- **Thematic Coherence:** The baseline loses its thematic thread within 3 to 4 lines. The DPO model typically sustains the original theme for 6 to 9 lines before drifting.
- **Vocabulary:** The DPO model limits hallucinatory/invented words (12 down from 19 in baseline), producing plausible archaisms instead of noise.

Paraphrase Identification: Our deep dive into edge cases revealed semantic ambiguity on labels. Often human annotators struggle on these boundary cases, which explains the constrained mathematical improvement ceiling post-SFT.

Conclusions

- DPO brings measurable, observable value beyond its originally demonstrated open-ended text domains, proving effective for highly constrained creative tasks and offering marginal gains in discriminative classification.
- Our sonnet generation findings expose a critical gap between automatic evaluations (e.g. chrF) and genuine stylistic quality in text generation. The true improvements of preference alignment are often structurally profound but formally hard to quantify.

References